



HCI- Multimodal Systems

MODELLING NONVERBAL BEHAVIORS PART 2

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Let's delve into some of the
(popular) **nonverbal cues** and
discuss them for **affective
computing**

FACIAL EXPRESSIONS

- How to measure and compare two (facial) expressions?



beautiful smile



bright smile



smart smile

- How to show that two expressions by 2 different persons are “the same”?
- How can the machine classify facial expressions?
- Can we describe a facial expression in a more objective manner?



FACIAL ACTION CODING SYSTEM (FACS)

- By Paul Ekman and Wallace Friesen, 1978:
- Allows expert coders to manually measure facial expressions by breaking them down into component movements of individual facial muscles (i.e., action units)
- Inspired by human anatomy
- Encodes any visible change on the face (muscle deformation, apparition of wrinkles/bulges, folds), and secondary movements (propagation of muscle contraction)
- Taxonomizes human facial movements by their appearance on the face



FACIAL ACTION CODING SYSTEM (FACS)

- FACS can be applied by humans (certified annotators).
- Can be used to describe any facial expressions (not only emotions)
- Using FACS, nearly any anatomically possible facial expression is coded by deconstructing it into specific **Action Units (AU)** and their temporal segments producing the expression.
- AUs are independent of any interpretation and can be used for any higher-order decision-making process, e.g., recognition of emotions.



FACIAL ACTION CODING SYSTEM (FACS)

- FACS include:
- **Action Units (AUs)**: the fundamental actions of individual muscles or groups of muscles
 - Face: 46 AUs
 - Head: 14 AUs
 - Gaze: 11 AUs
- **Intensities** of AUs are annotated by appending a letter (A to E) to the Action Unit number, meaning that intensity can be expressed in 5 levels scale:
 - A: Trace
 - B: Slight
 - C: Marked or Pronounced
 - D: Severe or Extreme
 - E: Maximum
- certain AUs can be Left (L) or Right (R)

Facial Action Coding System



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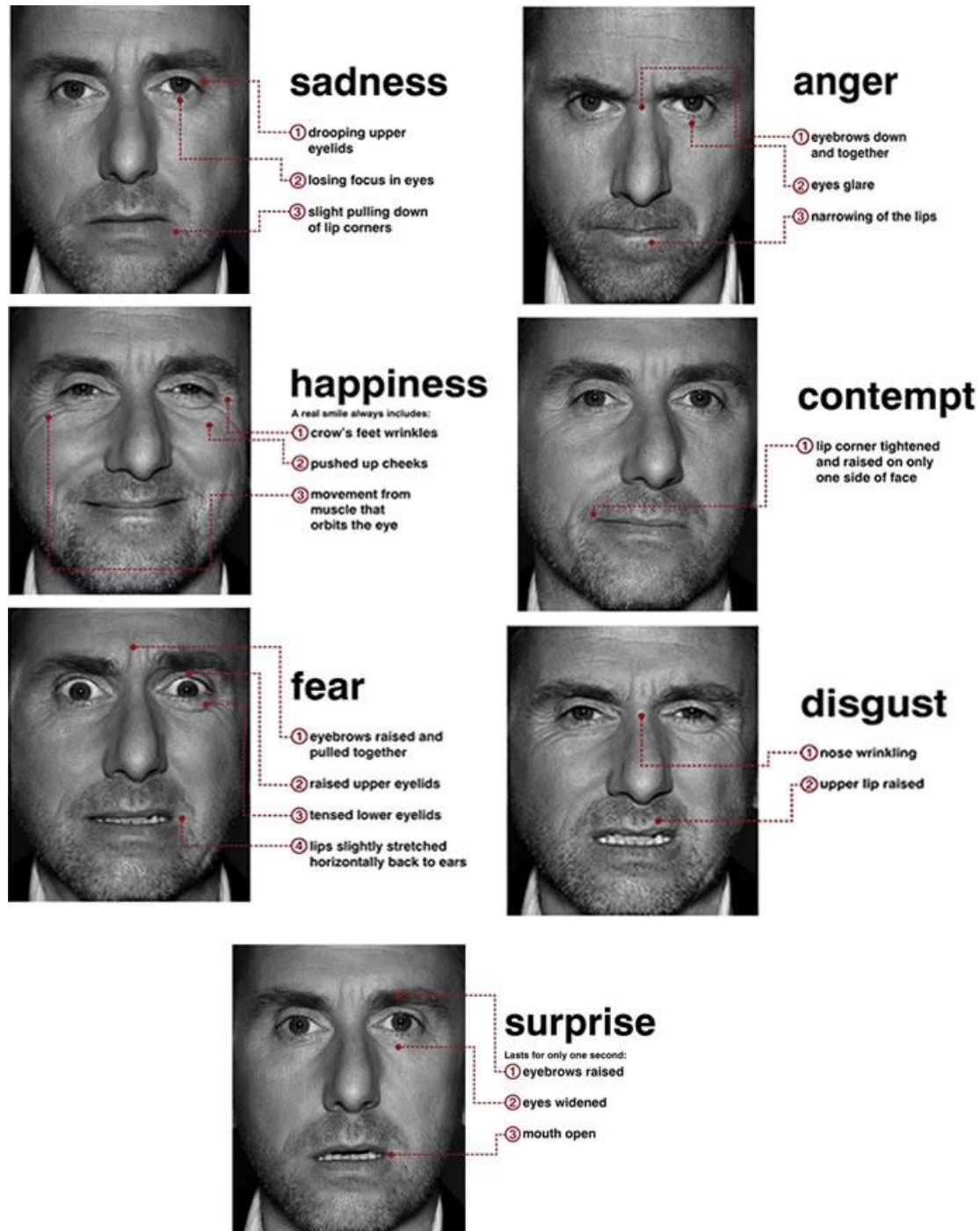


<https://www.youtube.com/watch?v=-G7IRRydpVA>

Upper Face Action Units					
AU 1	AU 2	AU 4	AU 5	AU 6	AU 7
					
Inner Brow Raiser	Outer Brow Raiser	Brow Lowerer	Upper Lid Raiser	Cheek Raiser	Lid Tightener
*AU 41	*AU 42	*AU 43	AU 44	AU 45	AU 46
					
Lid Droop	Slit	Eyes Closed	Squint	Blink	Wink
Lower Face Action Units					
AU 9	AU 10	AU 11	AU 12	AU 13	AU 14
					
Nose Wrinkler	Upper Lip Raiser	Nasolabial Deepener	Lip Corner Puller	Cheek Puffer	Dimpler
AU 15	AU 16	AU 17	AU 18	AU 20	AU 22
					
Lip Corner Depressor	Lower Lip Depressor	Chin Raiser	Lip Pucker	Lip Stretcher	Lip Funneler
AU 23	AU 24	*AU 25	*AU 26	*AU 27	AU 28
					
Lip Tightener	Lip Pressor	Lips Part	Jaw Drop	Mouth Stretch	Lip Suck

It is possible to have combinations of Aus.
But, some combinations are not “intuitive”...

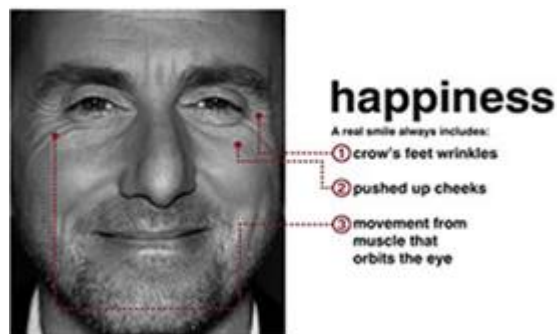
<i>NEUTRAL</i>	AU 1	AU 2	AU 4	AU 5
				
Eyes, brow, and cheek are relaxed.	Inner portion of the brows is raised.	Outer portion of the brows is raised.	Brows lowered and drawn together	Upper eyelids are raised.
AU 6	AU 7	AU 1+2	AU 1+4	AU 4+5
				
Cheeks are raised.	Lower eyelids are raised.	Inner and outer portions of the brows are raised.	Medial portion of the brows is raised and pulled together.	Brows lowered and drawn together and upper eyelids are raised.
AU 1+2+4	AU 1+2+5	AU 1+6	AU 6+7	AU 1+2+5+6+7
				
Brows are pulled together and upward.	Brows and upper eyelids are raised.	Inner portion of brows and cheeks are raised.	Lower eyelids cheeks are raised.	Brows, eyelids, and cheeks are raised.



Examples(!) of expressions in FACS notation

Emotion ◆	Action units ◆
Happiness	6+12
Sadness	1+4+15
Surprise	1+2+5B+26
Fear	1+2+4+5+7+20+26
Anger	4+5+7+23
Disgust	9+15+17
Contempt	R12A+R14A

A prototypical “happy face” is a combination of...



AU 6 + 7 which are muscle activities around the eyes

AU 12 – which is a zygomatic major activity – the most important element of the smile

They can be accompanied by AU25 and AU26

Upper Face Action Units					
AU 1	AU 2	AU 4	AU 5	AU 6	AU 7
					
Inner Brow Raiser	Outer Brow Raiser	Brow Lowerer	Upper Lip Corner Puller	Cheek Raiser	Lid Tightener
*AU 41	*AU 42	*AU 43	AU 44	AU 45	AU 46
					
Lid Droop	Slit	Eyes Closed	Squint	Blink	Wink
Lower Face Action Units					
AU 9	AU 10	AU 11	AU 12	AU 13	AU 14
					
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AU 23	AU 24	*AU 25	*AU 26	*AU 27	AU 28
					
Lip Tightener	Lip Pressor	Lips Part	Jaw Drop	Mouth Stretch	Lip Suck

EMOTIONS

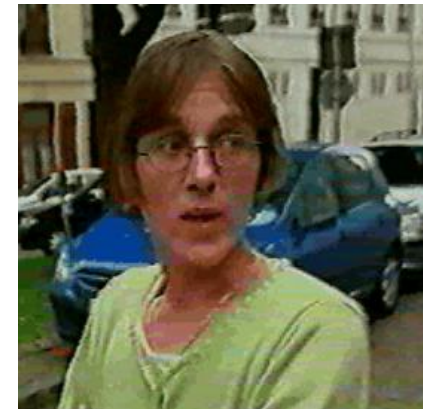


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- **Discrete Approach (Ekman)**
 - Anger, fear, sadness, happiness, disgust, surprise
 - Universally recognized (Ekman)
 - basic emotion = family of related states (Ekman 75)
- **Real-life emotions**
 - Are often complex and involve several emotions
 - Theories (Ekman 75, Plutchik 80, Ekman 92, Scherer 84)
 - Lost luggage study (Scherer 98)
 - TV interviews : Belfast & EMoTV corpora (Douglas-Cowie et al. 2005)

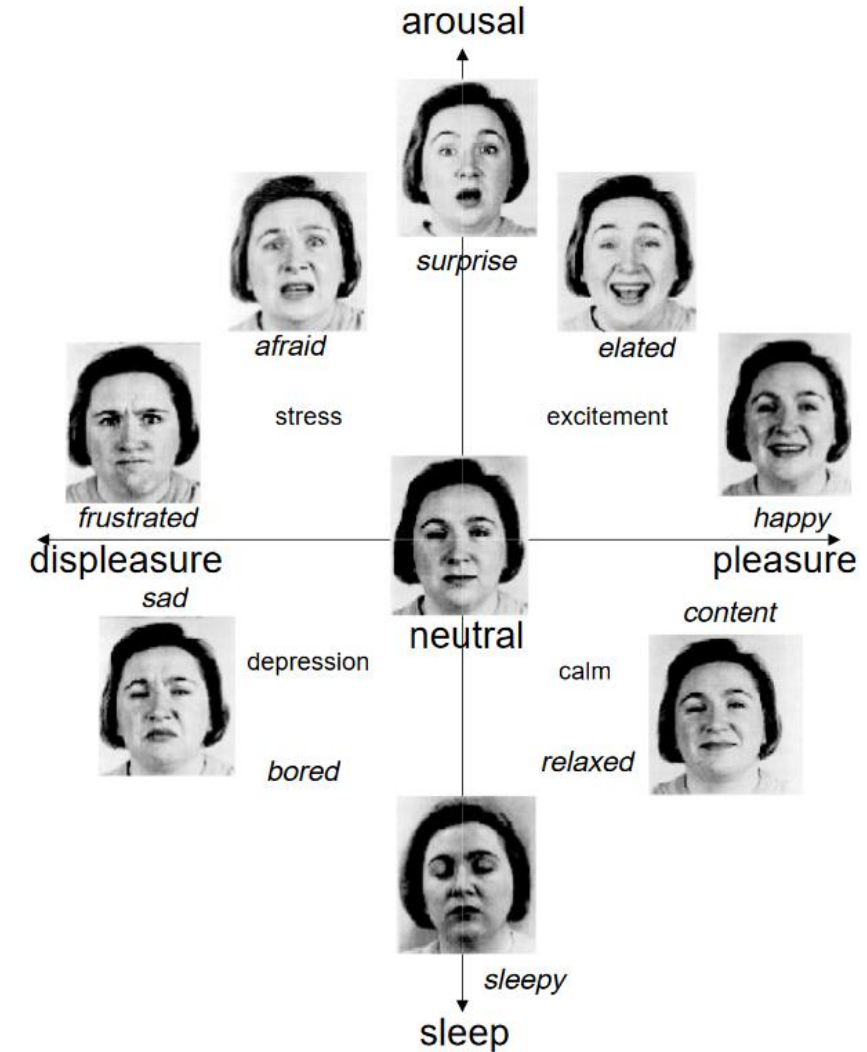


(Ekman 75)



EMOTIONS

- If there are an infinite number of emotions, then there can be an infinite number of expressions
- **Russell 2D model** is based on this idea such that
 - some easily distinguishable expressions are close to each other in this model.
 - Emotions are defined based on two fundamental dimensions: **valence** and **arousal**.
- **Valence**: measures how positive or negative an emotion is.
 - E.g., happiness has a positive valence, while sadness has a negative valence.
- **Arousal**: intensity associated with an emotion.
 - E.g., excitement represents high arousal, while calmness represents low arousal.



Breazeal, 2003

EMOTIONS



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- Can we infer all emotions in all circumstances from human facial movements?
- Are facial expressions a comprehensive solution for understanding emotions?



Emotional Expressions Reconsidered: Challenges to Inferring Emotion From Human Facial Movements

Psychological Science in the
Public Interest
2019, Vol. 20(1) 1–68
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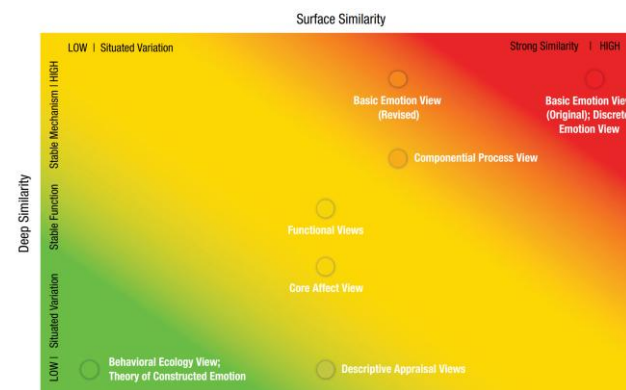


Fig. 1. Explanatory frameworks guiding the science of emotion: the nature of emotion categories and their concepts. The information in the figure is plotted along two dimensions. The horizontal dimension represents hypotheses about the similarities in surface features shared by instances of the same emotion category (e.g., the facial movements that express instances of the same emotion category). The vertical dimension represents hypotheses about the similarities in the mechanisms that cause instances of the same emotion category (e.g., the neural circuits or assemblies that cause instances of the same emotion category). The colors represent the type of emotion categories proposed in each theoretical framework. Approaches in the green area describe ad hoc, abstract categories; those in the yellow area describe prototype or theory-based categories, and those in the red area describe natural-kind categories.

<https://journals.sagepub.com/doi/pdf/10.1177/1529100619832930>

Is it soo.... easy?



In real life faces are rarely encountered in isolation

Context



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a



Context



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b



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Context



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C



Context



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d



Context



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a



b



c



d



Context: categorical approach

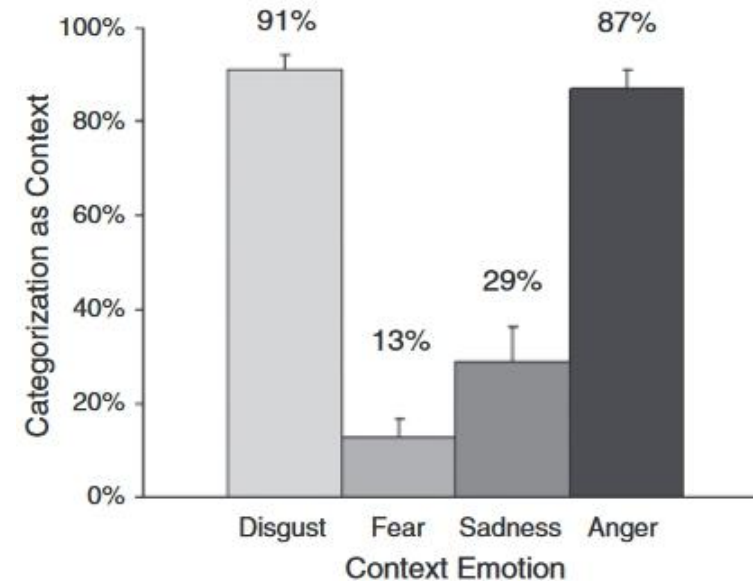
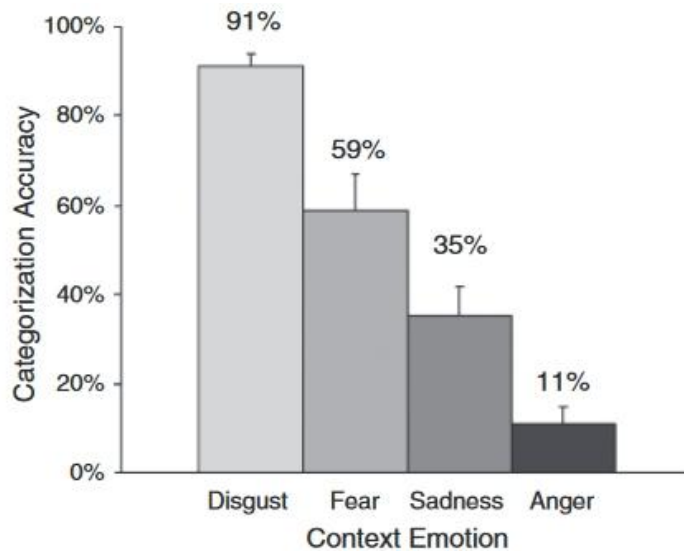


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a



b



the percentage
of times the
face was
categorized as
“disgust”

the percentage of
times the face
was categorized
as expressing the
context emotion

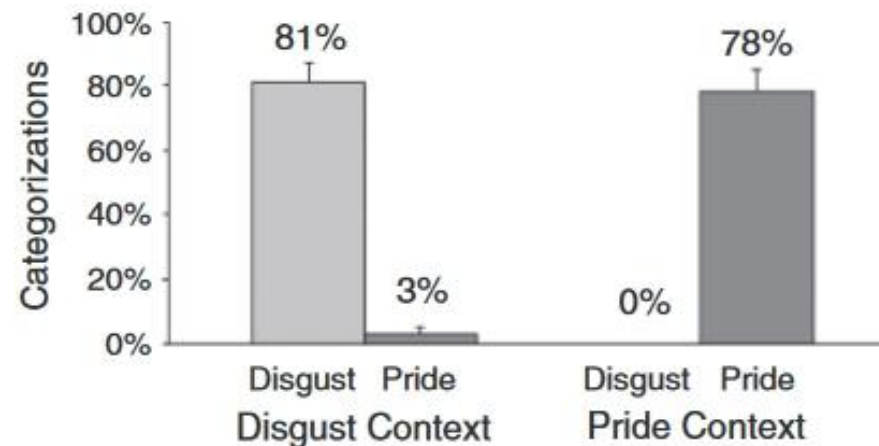
Context: dimensional approach



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“Context” is understood broadly:

- gestures and postures
- object manipulations



Switching between
positive and negative
valence context



EMOTIONS: TAKE HOME MESSAGES

- Widely used theories of emotion claim that basic facial expressions signal
 - discrete emotions, or affective dimensions
- In both cases, the information is thought to be directly “read out” from the face
- In contrast, studies show that:
 - identical facial configurations convey different emotions (and dimensional values) depending on the (affective) context
 - the perception of basic facial expressions is not context-invariant and can be altered by context
- Similarity between the expressions (target label, perceived label) matters. Labels of similar expressions are more easily “exchanged”
- In real life, faces are rarely encountered in isolation, and the context in which they appear is very informative
- **WHICH “EMOTION RECOGNITION” SOFTWARE USES THE CONTEXT INFO?**



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Gestures & Full-body Movement

GESTURE



- “A natural gesture means the types of gestures **spontaneously** generated by a person telling a story, speaking in public, or holding a **conversation**” (Cassel et al., 1990)
- “A Movement of the arms and hands which are closely **synchronized with the flow of speech**” (McNeill, 1992)
- “A movement of one’s body that conveys meaning to oneself or to a partner in communication” (Hummels et al., 1998)
- “A movement of the body that contains **information**” (Kurtenbach and Hulteen, 1990).

MCNEILL'S GESTURE TAXONOMY



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- Highlights how gestures are integrated with speech and convey meaning.
- **Iconic Gestures:** These gestures visually represent the content of the speech.
 - Ex: Describing a tree and using your hands to outline its shape.
- **Metaphoric Gestures:** These gestures represent abstract concepts or ideas.
 - Ex: Using a sweeping hand motion to indicate the passage of time.
- **Deictic (Pointing) Gestures:** These gestures indicate objects, locations, or directions.
 - Ex: Pointing towards a place when giving directions.

MCNEILL'S GESTURE TAXONOMY



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- **Beat Gestures:** These gestures follow the rhythm of the speech but do not convey specific content.
 - Ex: Moving hands up and down in sync with the stress patterns of speech to emphasize a point.
- **Cohesive Gestures:** These gestures tie together parts of discourse, linking segments of speech to create coherence.
 - Ex: Using hand motions to indicate that two ideas are connected.

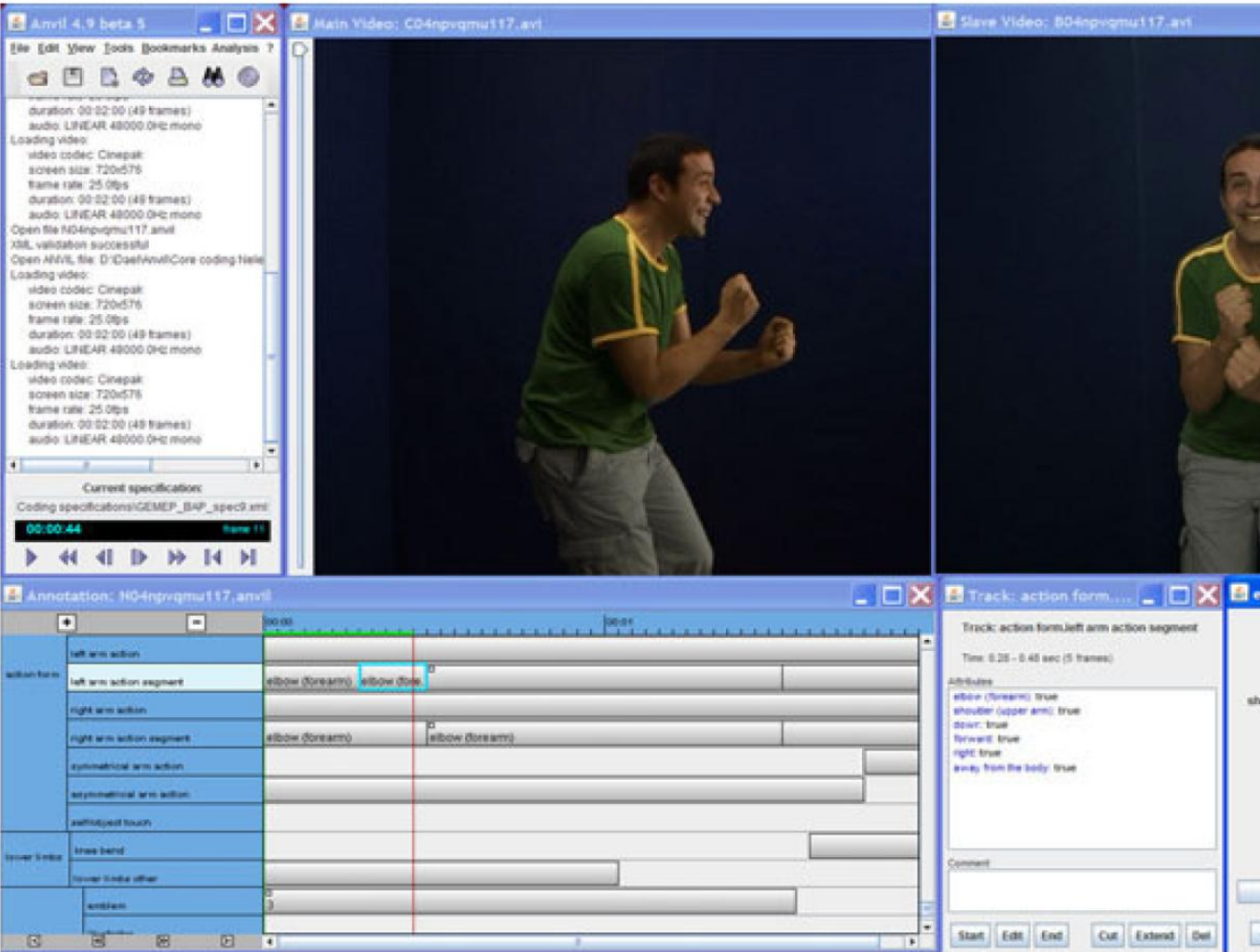
GESTURES & BODY MOVEMENTS



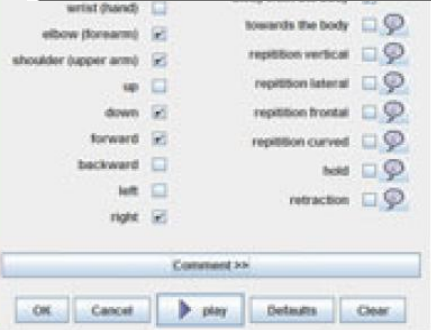
- Do we have some standards like FACS to describe the body movements?
 - No standard but there are some attempts.
- One example is: **Body Action and Posture Coding System (BAP)**: which focuses on visible body changes, rotations, etc. [Dael, et al. 2012]
 - 141 behaviors: describes the position and movement of body and body parts, e.g., fingers, head, arm.
 - Communicative intent of a gesture.

<i>Whole body posture</i>		
31	Whole body moves or leans towards a forward position	The whole body, including the lower part (hips and legs) moves or leans forward relative to the anatomical standard position
32	Whole body moves or leans towards a backward position	The whole body, including the lower part (hips and legs) moves or leans backward relative to the anatomical standard position
33	Whole body moves or leans towards the frontal middle position	The whole body, including the lower part (hips and legs) moves or leans forward or backward towards the standard anatomical position
34	Whole body moves or leans towards a left position	The whole body, including the lower part (hips and legs) moves or leans towards the left relative to the anatomical standard position
35	Whole body moves or leans towards a right position	The whole body, including the lower part (hips and legs) moves or leans towards the right relative to the anatomical standard position
36	Whole body moves or leans towards the lateral middle position	The whole body, including the lower part (hips and legs) moves or leans left or right towards the standard anatomical position
<i>Arms posture</i>		
37	Left arm at side	The left arm and hand hangs at the side of the body
38	Left arm held in front	The left hand and/or arm is held in front of the body
39	Left arm held behind back ^b	The left hand and/or arm is held behind the back
40	Left hand in pocket	The left hand (or a part of it) is in the pocket
41	Left hand at waist	The left hand rests on the hip
42	Left hand at back of head/neck ^a	The left hand is held behind the back of the head or neck

BAP was utilized in performance arts and animation to create more lifelike characters by accurately representing human body language.



Nodes	
LA side Container	Left arm at side. The left arm and hand hangs at the side of the body.
LA side-T Container	Transition phase of left arm at side.
LA side-C Container	Configuration phase of left arm at side.
LA front Container	Left arm held in front. The left hand and/or arm is held in front of the body.
LA front-T Container	Transition phase of left arm held in front.
LA front-C Container	Configuration phase of left arm held in front.
LA back Container	Left arm held behind back. The left hand and/or arm is held behind the back.
LA back-T Container	Transition phase of left arm held behind back.
LA back-C Container	Configuration phase of left arm held behind back.
LH pocket Container	Left hand in pocket. The left hand (or a part of it) is in the pocket.
LH pocket-T Container	Transition phase of left hand in pocket.
LH pocket-C Container	Configuration phase of left hand in pocket.
LH waist Container	Left hand at waist. The left hand rests on the hip.
LH waist-T Container	Transition phase of left hand at waist.
LH waist-C Container	Configuration phase of left hand at waist.
LH neck Container	Left hand at back of head/neck. The left hand is held behind the back of the head or neck.
LH neck-T Container	Transition phase of left hand at back of head/neck.
LH neck-C Container	Configuration phase of left hand at back of head/neck.
RA side Container	Right arm at side. The right arm and hand hangs at the side of the body.



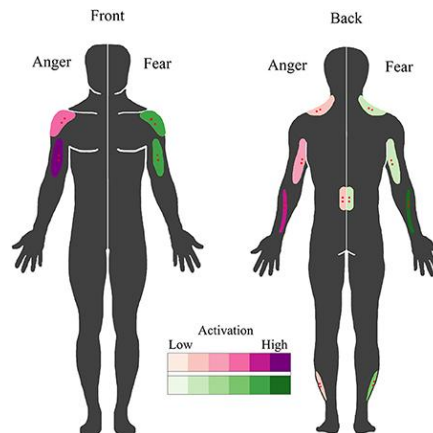
GESTURES & BODY MOVEMENTS



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- Another example: **Body Action Coding System (BACS)**
- BACS categorizes and codes a wide range of body actions, including gestures, postures, and movements.
- Combinations of muscles are used for emotional bodily expressions.
- Assessing whether these muscles also automatically respond to the perception of emotion.
- **Electromyography (EMG)** of different body parts muscles measured during active expression and passive viewing of emotional bodily expressions.

Anger: pink
Fear: green



Huis in 't Veld et al. 2014

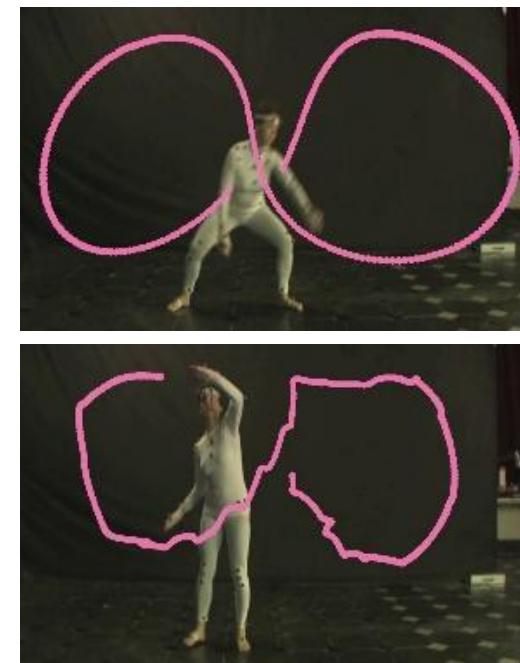
AFFECT EMBLEMS & ADAPTORS

- **Affect emblems** are nonverbal gestures that convey specific emotional states or feelings without words.
- Ex: thumbs up (approval), a heart shape made with hands (love, affection), crossed arms with a frown (displeasure or defensiveness)
 - Do not imply that the sender is actually experiencing an emotion.
 - Can be culture-dependent. e.g., the “hand purse” gesture (figure on top) represents fear in France but not in Italy.
- **Adaptors**: self-touching behaviors or actions that people use to manage emotions, relieve stress, or regulate physical and emotional states.
 - E.g., biting nails, tapping fingers, touching face or hair



EXPRESSIVE GESTURES

- Gestures not only denote things or support speech, but also:
 - communicate affect and emotion
 - are social signals (e.g., cohesion in a group of humans)
- Expressive gesture is conceived as a movement of the body that contains and conveys expressive information, e.g., information concerning affect and emotion
- Expressive gesture encodes:
 - content information (the ‘*What is communicating*’)
 - expressive information (the ‘*How it is communicating*’)
- Behavior expressivity (or expressive quality) refers to the manner of execution of the behavior



Piana et al., 2015



The same action
and object but
different “How”



What is the
difference?

How would you
call them?



The same expressive content (“how”) and different gestures

Task 2. Vitality forms recognition

Next, you will see a short extract from the data collection session.

In the video, two different actions will be performed gently and rudely.

EXPRESSIVE CONTENT/QUALITY



- **Expressive gestures** are nonverbal actions that communicate more than just factual information; they also convey the person's internal states, such as emotions, moods, attitudes, and social intentions.
- These gestures do not just show what someone is doing; they show **how** someone is feeling while they are doing it. This is referred to as **expressive content** or **expressive quality**.
- Expressive content is different and often independent from the gesture's literal meaning
 - E.g., a raising hands expressing excitement or celebration as denotative meaning but if it is performed slowly, it might convey a different emotional tone such as tiredness.



LABAN'S MOVEMENT ANALYSIS

- Does expressive quality have any formal description?
- **Laban's movement analysis** is a system developed to systematically describe and understand the broad characteristics of how humans move.
 - Instead of focusing on specific types of movement, like dance steps, athletic motions, or everyday actions, this analysis aims to capture the **fundamental aspects** of all human movement in a **generalized manner**.
 - The actions of the body parts over time are described by using a **notation system**.

LABAN'S MOVEMENT ANALYSIS



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- 4 components of **LMA**:
- **Body**: The body parts involved in the movement
- **Space**: Involves the direction, level, and pathway of the movement within space.
- **Effort**: Describes the dynamics or qualities of the movement, focusing on how the movement is performed.
- **Shape**: Refers to the forms or shapes that the body makes as it moves through space. E.g., how body is oriented or changing forms

LABAN'S MOVEMENT ANALYSIS: EFFORT



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- Effort category includes: space, time, weight, and flow.
- **Space:** direct vs. indirect
 - Direct: focused, precise, and has a clear and purposeful direction. E.g., pointing straight ahead.
 - Indirect: flexible, multi-directional, and adaptable, with less on a specific target. E.g., waving your hand loosely in different directions.
- **Time:** quick/sudden vs. sustained
 - Quick/sudden movement: performed with urgency and speed, often sharp and instantaneous. E.g., sudden jump
 - Sustained: slow, continuous, and drawn out over time. E.g., a slow, graceful arm wave

LABAN'S MOVEMENT ANALYSIS: EFFORT



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- **Weight:** Strong vs. light
 - Strong movement: uses a lot of force or power, conveying a sense of firmness or solidity. E.g., pushing a heavy object.
 - Light movement: gentle, delicate, and uses minimal force. E.g., gently brushing something away.
- **Flow: free vs. bound**
 - Free movement: fluid, ongoing, and less restricted, allowing for more freedom of motion. E.g., a carefree dance
 - Bound movement: controlled, contained, and more restrained, with a clear sense of stopping or holding back. E.g., carefully pouring a liquid.

LABAN'S MOVEMENT ANALYSIS



Table 5. An overview of discriminative body movements and postures based on *post-hoc* comparisons reported by Wallbot (1998)

Emotion	Upper body	Shoulders	Head	Body Parts Arms	Hands	Movement quality
	Erect	Hanging	Toward camera	Hanging	No specific activities	No extreme judgments
Cold anger Hot anger		Up		Stretched out frontal	Lateralized movements	High movement activity, expansive movements, high movement dynamics
Elated joy		Up	Backward	Stretched out frontal or upward	Opening/closing, many illustrators	High movement activity, expansive movement, high movement dynamics
Happiness Disgust	Collapsed	Backward or forward	Downward	Crossed in front of chest		Low movement dynamics Inexpansive movements
Contempt Sadness Despair Fear	Collapsed	Forward			Opening/closing Opening/closing, many manipulators	Low movement activity, low movement dynamics Expansive movements
Terror				Stretched out sideways		High movement activity
Shame Interest Pride	Collapsed		Downward Backward	Stretched out frontal Crossed in front of chest	Many self-manipulators Lateralized movements	
Boredom	Collapsed		Backward			Low movement activity, inexpansive movements, low movement dynamics

POSTURE



- Refers to the quasi-static configuration of the whole body
- Usually defined by **the position** and **orientation** of specific body parts
- Different approaches have been proposed
 - Mehrabian, 1972: **global descriptors**, e.g., relative orientation to the partner, closeness or openness, and symmetry.
 - E.g., arms crossed are closed, standing with arms relaxed at the sides is open. Standing evenly with weight on both feet vs. leaning to one side are matters of symmetry.

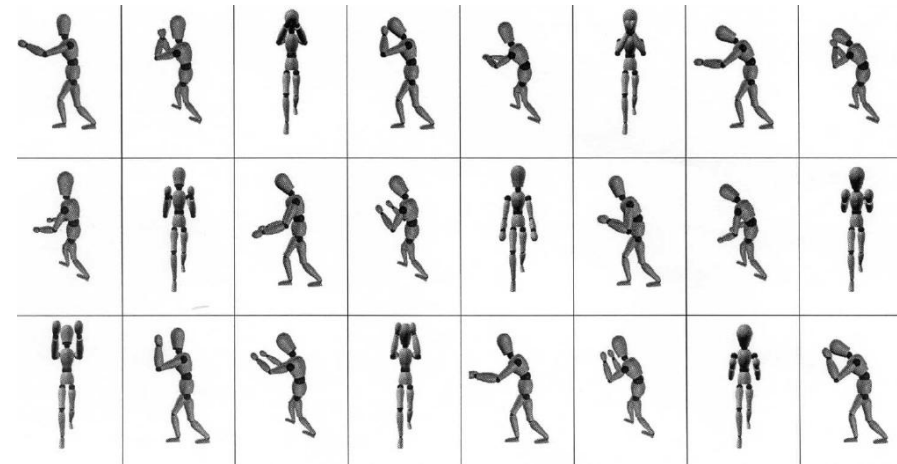
POSTURE



- Harrigan, 2005: presents a **coding system**, using descriptions:
 - **Trunk lean**: whether your torso is upright, leaning forward, or leaning backward
 - **Trunk orientation**: Whether your torso is facing directly towards something or someone or turned away.
 - **Arms and leg positions and configurations**: Describes how your arms and legs are positioned, such as having your hands in your pockets or crossing your legs.
- Gross, Crane, and Fredrickson, 2012: **absolute orientation** values regarding:
 - The **sagittal axis** refers to front-to-back movements (e.g., bending forward or backward), the **frontal axis** refers to side-to-side movements (e.g., tilting to the left or right), the **transverse axis** refers to rotational movements around a vertical axis (e.g., twisting to face left or right).

EMOTIONS IN POSTURES

- By Coulson (2004):
- 176 computer-generated postures of avatars from 3 viewing angles
- Manipulations: abdomen twist, chest, elbow, and head bend, shoulder swing...
- Perceptive study (a forced-decision task)
 - 6 labels (basic emotions)
 - 61 participants
- Results:
 - High agreement for some **anger** and **sadness**
 - No agreement for **disgust**



“Anger” postures



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Touch

TOUCH



- Relatively less studied compared to other nonverbal cues because
 - Data collection is difficult (i.e., intimacy, privacy)
 - In some cultures, touch is not welcome as a communication tool.
 - Touch sometimes associated with mobbing, sexual abuse, etc.
 - Sensor systems are still underdevelopment
 - Only recently, interesting devices were developed such as artificial skin, etc.
- Touch has not been often associated with states and emotions 😞
 - In fact several positive emotions can be expressed with touch



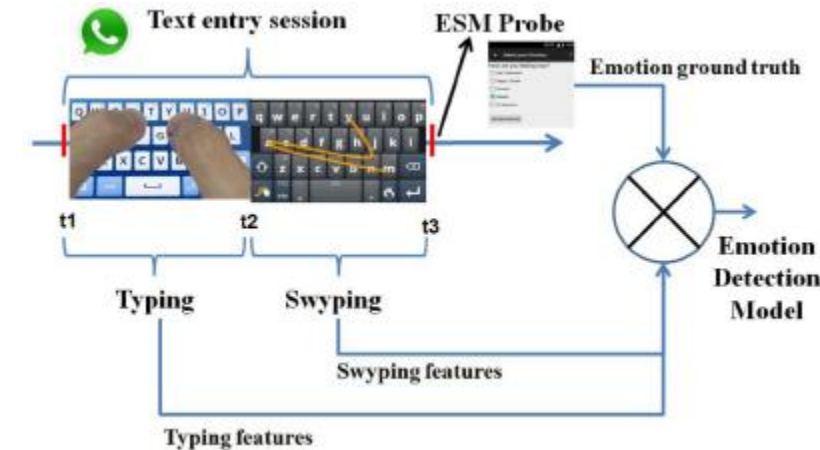
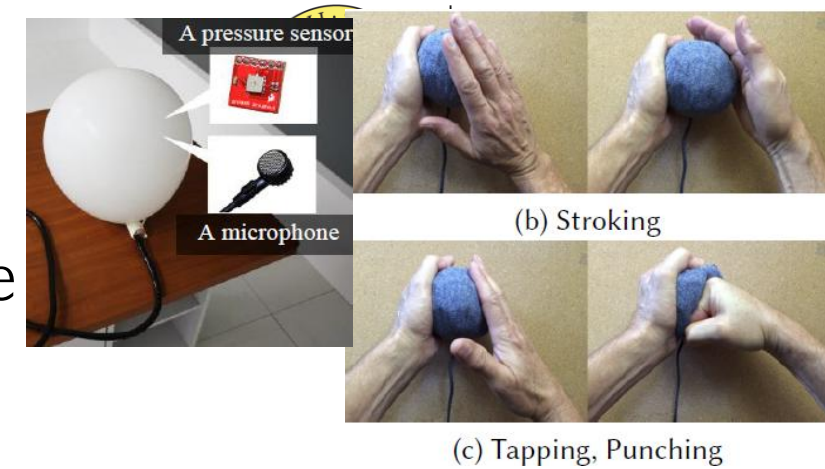
TOUCH AND EMOTIONS

Hertenstein et al. 2009:

- How accurately does the *person being touched to* decode the intended emotions without seeing it?
- Two strangers interact in a room where they are separated by a barrier, they do not see one another and they can reach each other through a hole in the barrier
- One person touched the other's arm to convey different emotions
 - 6 Ekmanian emotions + 3 prosocial (love, gratitude, sympathy) + 3 self-focused (embarrassment, pride and envy)
 - 106 dyads, balanced genders in dyads

TOUCH- RELATED WORKS

- Researchers were able to automatically classify affective states from, e.g., **soft fidget objects** by applying ML/DL (*Nakajima et al., 2013* *Shapiro et al., 2019*), **keyboard tapping and touchpad patterns** of mobile devices (Ghosh et al. 2019)
- **Finger stroke features** during gameplay on a mobile device (Gao et al. 2012)
 - four emotional states
 - Applies discriminant analysis:
 - Results: pressure-related features distinguish frustration from the other states, stroke length-related features differentiate boredom from a relaxed state, and classification accuracy reaches up to 77%





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Physiological Signals



PHYSIOLOGICAL SIGNALS AND EMOTIONS

- Good at detecting felt («real») emotional states.
- Sensors are now small and portable.
- Usually, a combination of different sensors is needed.
- Only a few emotions have been studied so far.
- The same physiological reaction can have different origins:
 - Heart Rate Variability (HRV) decreases with negative emotions (e.g., sadness),
 - HRV decreases with mental effort and conscious control.
- High variability between people (difficult to build a single model)
- Technical and ethical constraints: a) long-term use have some health consequences? (e.g., skin problems), b) privacy in-the-wild setting, c) differences in-the-lab and in-the-wild setting

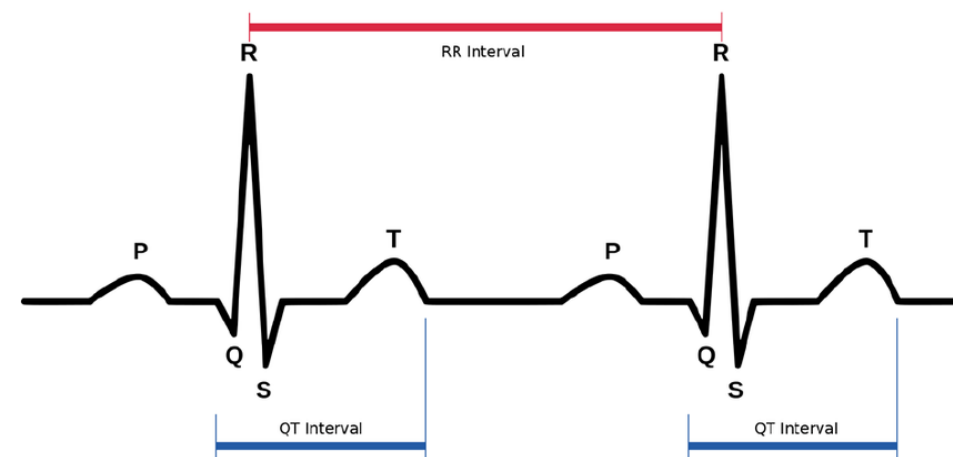
PHYSIOLOGICAL MEASURES



Full name (abbreviation)	Measure of	Quantification of
Electrocardiography (ECG)	Voltage changes	HR, IBI, PWA, TWA, HRV, RR (spectral)
Impedance cardiography (ICG)	Impedance changes	HR, IBI, PEP, LVET, SV, CO, RR
Photoplethysmography (PPG)	Volume changes	HR, PA, PTT (in combination with ECG)
Sphygmomanography	Pressure changes	HR, SBP, DBP, MAP, TPR (in combination with ECG)
Temperature (TMP)	Temperature changes	FT
Respiratory inductive plethysmography (RIP)	Volume changes	w
Impedance pneumography (IPG)	Impedance changes	RR, Ttot, Vt, Ti/Ttot, Vt/Ti
Phonspirometry	Sound changes	RR, Ttot, Vt, Ti/Ttot, Vt/Ti
Temperature	Temperature changes	RR
Capnography	Changes in blood gas saturation	pCO ₂
Pulse oximetry	Changes in blood gas saturation	SO ₂
Electrodermography (EDG)	Conductance changes	SCL, NS-SCR, SRA

PHYSIOLOGICAL SIGNALS AND EMOTIONS

- **Heart rate (HR) & heart rate variability (HRV):** most commonly used in affective computing
 - Stress: HR increases, relaxation: HR decreases
 - Many features can be extracted from periods of acceleration and deceleration in the HR, such as the mean difference over baseline, and the amount of time spent in acceleration vs. deceleration and the magnitude of the acceleration or deceleration, difference between the max. and min. normal R.-R interval
 - Used for: Indicator of fear (Levenson, 1992), panic (Hofmann & Barlow, 1996), anger (Levenson, 1992) and appreciation (McCraty et al., 1995)





A Brief Intro to Emotion Theories

DEFINITION OF EMOTION(S)

- No **ONE** widely accepted definition of emotion
- Emotion is....
 - “...an episode of interrelated, synchronized changes in the states of all or most of the five organismic subsystems in response to the evaluation of an external or internal stimulus event as relevant to major concerns of the organism”*
- Five components
 - perceptive-cognitive component: influences or influenced by thinking
 - neurophysiological component: e.g. arousal, heart-rate, sweating...
 - motivational component (action tendency): e.g., to flight
 - motor expression component: facial expressions, posture, vocal features
 - subjective feeling: conscious awareness being in an emotional state



AFFECT RELATED PHENOMENA

- Moods, interpersonal stances, preferences/attitudes, personality...
- Significant differences:
 - Emotions between 1 and 10 seconds
 - Personality traits may change only very slowly
 - Moods are not focused on the particular event



MAIN EMOTION THEORIES

- **Discrete:** a limited number of fundamental emotions (called “primary” or “basic”) characterized by a specific expression, and physiological response
- **Dimensional:** emotion is a point in *n-dimensional* space; the number of emotions is infinite
- **Appraisal:** emotion arises from a series of sequential evaluation checks of the surrounding stimuli; each step is linked to a facial response.
 - For instance, if you perceive a situation as a threat, you might feel fear. If you view the same situation as a challenge you can overcome, you might feel excitement or motivation instead.

DISCRETE EMOTION THEORIES



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- For example: Ekman, Izard, Tomkins
- There are a (small) number of “basic” emotions such as anger, fear, joy, sadness, disgust, and surprise
- Emotions are determined by neuromotor mechanisms
- Each emotion is associated with an “affect” program producing a prototypical profile including a specific action tendency, physiological response pattern, motor expression (e.g., facial expressions), and feeling state
- The “affect” program is activated by suitable perceptions or appraisals
- Each emotion has its evolutionary function
- More complex emotions are a mix of basic ones

SOME DISCRETE MODELS



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- **Ekman:** sadness, happiness, anger, fear, disgust, and surprise, (contempt added later on)



- **Tomkins:** excitement, joy, surprise, distress, anger, fear, shame, dissmell (string aversion to odors), disgust

<http://www.tomkins.org/what-tomkins-said/introduction/nine-affects-present-at-birth-combine-to-form-emotion-mood-and-personality/>

- **Izard & Malatesta:** happiness, surprise, sadness, anger, disgust, contempt, fear

DISCRETE MODELS: PROBLEMS



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- (Intuition) There are more emotions than on any list.
- Cultural differences can affect how emotions are expressed and perceived.
- Emotions are often based on personal evaluations or appraisals of events. Different individuals might evaluate the same event in various ways, leading to different emotional responses. For instance, one person might feel joy at receiving a promotion, while another might feel stress or anxiety due to increased responsibilities.

DISCRETE MODELS: PROBLEMS



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- Emotional experiences and concepts can be expressed differently in various languages, and some emotions may have no direct translation. For example, "schadenfreude" (German) refers to the pleasure derived from someone else's misfortune, and "fiero" (Italian) describes a feeling of pride and triumph, while "orgoglioso" (also Italian) translates to pride in a different context.

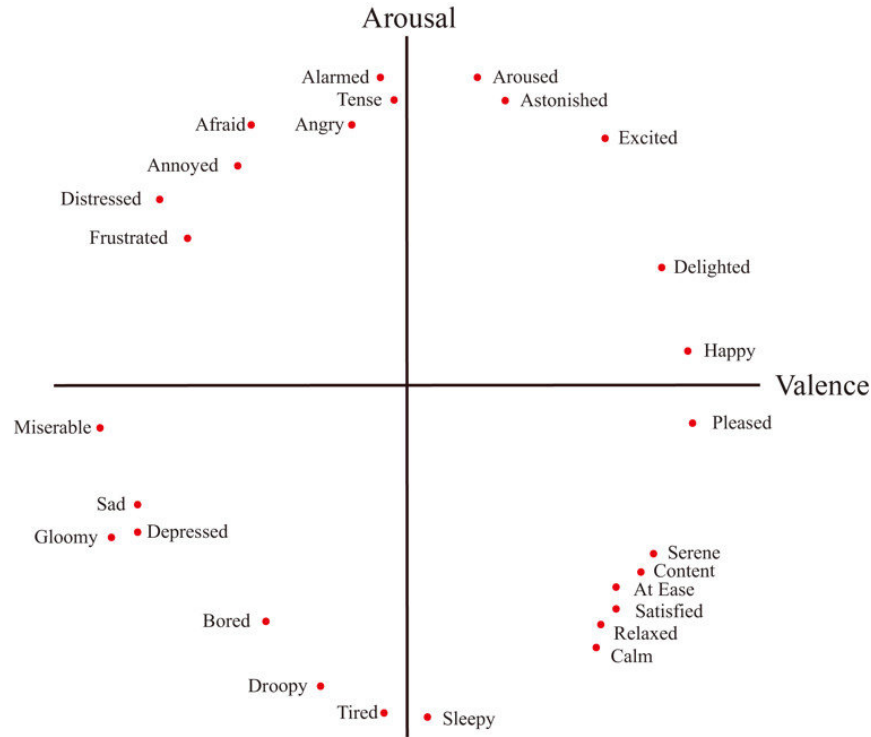
DIMENSIONAL MODELS



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- Emotions are points in a **continuous multidimensional space** where each dimension is a fundamental property against which all emotions are evaluated
- Dimensional theories propose that there is a link between an emotion, a facial expression (or its components), and these dimensions
- Problem: some “well-known” and distinct emotions are close to each other e.g., anger and fear

TWO DIMENSIONAL MODELS



Russell (1980): Two dimensions

- Valence (pleasant-unpleasant dimension)
- Arousal (high/low)
- Each emotion is a linear combination of these two dimensions
 - E.g., happiness is the product of strong activation in the neural systems associated with *positive valence* and moderate activation in the neural systems associated with *arousal*

Positive Valence + High Arousal: Excitement, euphoria

Positive Valence + Low Arousal: Contentment, relaxation

Negative Valence + High Arousal: Anger, fear

Negative Valence + Low Arousal: Sadness, depression

THREE DIMENSIONAL MODELS

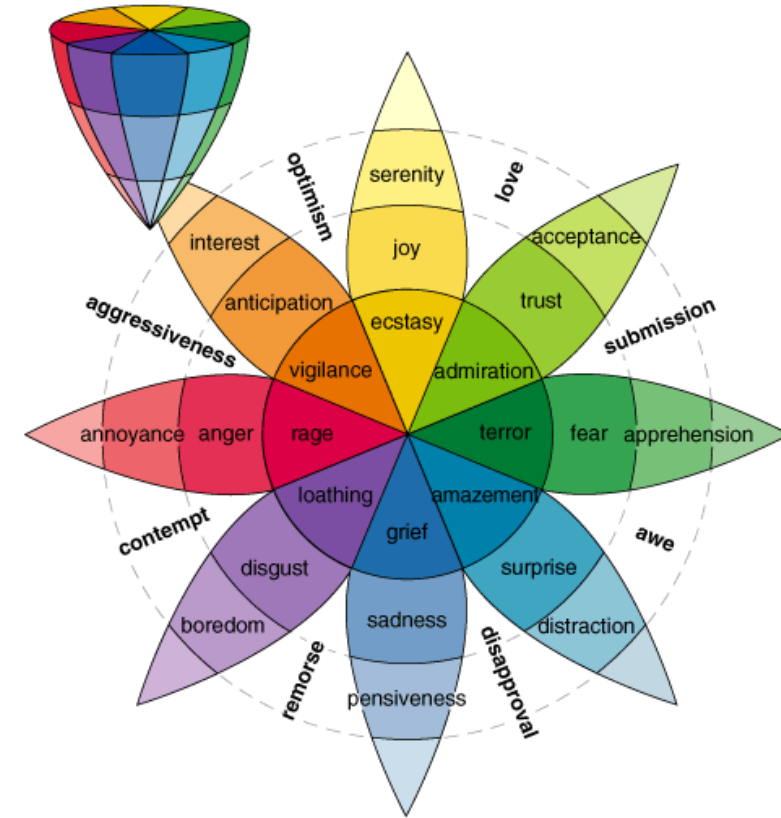
- The Russell's model is often used but other researchers considered other important dimensions.
- The very early work by Wundt (1987) proposed 3-dimensions.
 - *On the basis of quality we may distinguish certain fundamental emotional forms corresponding to the chief affective directions distinguished before. This gives us **pleasurable** and **unpleasurable**, **exciting** and **depressing**, **straining** and **relaxing** emotions.*
- **Pleasure/unpleasure** corresponds to valence, the other two dimensions are new, and only partially correspond to arousal.

MEANWHILE THE EMOTION WHEEL BY PLUTCHIK (1980)



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- This is called a wheel but it is 3D and it is a cone in reality.
- Emotions that are similar to each other are placed next to each other. Just as adjacent colors on a color wheel blend smoothly into each other, adjacent emotions on the wheel share common characteristics and can easily blend into one another.
 - For example, “joy” and “trust” are next to each other, and they might combine to form the emotion of “love.”

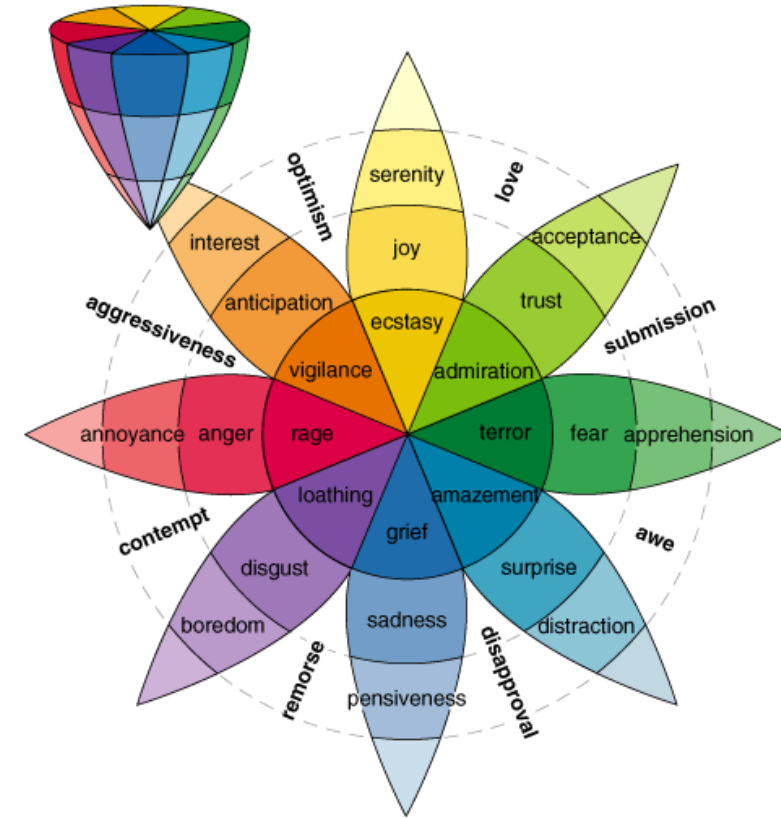


MEANWHILE THE EMOTION WHEEL BY PLUTCHIK (1980)



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- Emotions that are opposites are placed directly across from each other on the wheel, at a 180-degree angle.
 - For instance, “joy” and “sadness” are opposites, reflecting how they represent contrasting emotional states.
- **Intensity:** Basic emotions at the center (*rage—terror, vigilance—amazement, ecstasy—grief and admiration—loathing*) can have more intense versions, while those toward the outer edges are less intense.





APPRAISAL THEORIES

- Emotion arises from a **subjective interpretation** of the person-relationship environment
 - Individual judgment concerning the relationships between events, beliefs, desires, intentions
- Series of judgments (appraisals) of how some event impacts a person's beliefs, desires, and intentions
- Evaluation is “person dependent”
- Emphasis on the connection between emotion and cognition
- Several models exist, e.g., Antoniou, and Jose (1996), Roseman (2001), Lazarus (1991), Scherer (2001)
 - Main differences concern the number of list of appraisals



IRA ROSEMAN'S THEORY (1984)

- Five appraisals that can produce **14 discrete emotions**
- Appraisals are:
 - Motivational state (appetitive, aversive),
 - **Appetitive**: E.g., you receive a promotion at work. This is a desirable outcome because it advances your career and brings potential rewards. The emotion generated might be **joy** or **pride**.
 - **Aversive**: E.g., you fail an important exam. This is an undesirable outcome as it threatens your academic progress. The emotion generated might be **sadness** or **disappointment**.

IRA ROSEMAN'S THEORY (1984)



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- Situational state (motive-consistent, motive-inconsistent),
 - **Motive-consistent:** You achieve a personal goal, like running a marathon you've trained for. This aligns with your long-term aspirations and desires. The emotion experienced might be **satisfaction** or **euphoria**.
 - **Motive-inconsistent:** You lose a significant client for your business. This conflicts with your goal of growing your client base. The emotion experienced might be **frustration** or **anger**.
- Probability (certain, uncertain, unknown),
 - **Certain:** You know you're getting a bonus because it was confirmed by your manager. The certainty of this outcome might lead to **anticipation** or **excitement**.



IRA ROSEMAN'S THEORY (1984)

- Probability (certain, uncertain, unknown),
- **Uncertain:** You've applied for a prestigious grant but haven't heard back yet. The uncertainty of whether you will be awarded the grant might lead to **anxiety** or **hope**.
- **Unknown:** You're involved in a surprise event with no information about what will happen next. The complete lack of information might generate **curiosity** or **confusion**.
- Power (strong, weak),
 - **Strong:** You are leading a successful project and have full control over its direction. The perception of strong influence and control might lead to **pride** or **confidence**.

IRA ROSEMAN'S THEORY (1984)



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- Power (strong, weak):
 - **Weak:** You are facing a challenging situation at work with limited resources and little authority to make changes. The perception of weak influence might lead to **helplessness** or **frustration**.
- Agency (self-caused vs. other caused):
 - **Self-caused:** You forgot to prepare for an important presentation due to poor time management. The emotion might be **guilt** or **regret** because you view the outcome as a result of your own actions.
 - **Other-Caused:** A colleague takes credit for your work. The emotion might be **anger** or **resentment** because you perceive the event as being caused by someone else's actions.

IRA ROSEMAN'S THEORY (1984)



		Motive-Consistent		Motive-Inconsistent		← <i>situational state</i> ← <i>motivational state</i>
		Appetitive	Aversive	Appetitive	Aversive	
Circumstance-Caused	Unknown	Surprise				
	Uncertain	Hope		Fear		
	Certain	Joy	Relief	Sorrow	Discomfort, Disgust	Weak
	Uncertain	Hope		Frustration		Strong
	Certain	Joy	Relief			
Other-Caused	Uncertain	Liking		Disliking		Weak
	Certain			Anger		Strong
	Uncertain					
	Certain					
Self-Caused	Uncertain	Pride		Shame, Guilt		Weak
	Certain			Regret		Strong
	Uncertain					
	Certain					

↑
agency

↑
probability

↑
power



That's all

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